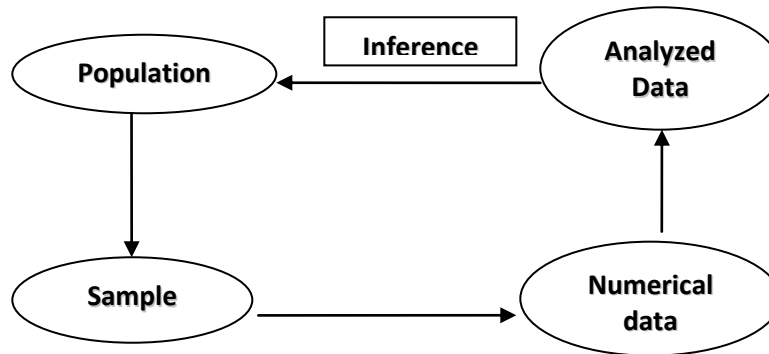


CHAPTER TWO STATISTICAL ESTIMATION

2.1. Basic Concepts

Statistical Inference is the process of making judgment about a population based on sampling properties. An important aspect of statistical inference is using **estimates** to approximate the value of an unknown population parameter. This chapter will study different kinds of estimator and lay the foundations for making statistical inference about the population mean and proportion.



Definitions

Confidence Interval: An interval estimate with a specific level of confidence

Confidence Level: The probability that the interval estimate will contain the parameter.

Consistent Estimator: An estimator which gets closer to the value of the parameter as the sample size increases.

Degrees of Freedom: The number of data values which are allowed to vary once a statistic has been determined.

Estimator: A sample statistic which is used to estimate a population parameter. It must be unbiased, consistent, and relatively efficient.

Estimate: Is the different possible values which an estimator can assumes.

Interval Estimate: A range of values used to estimate a parameter.

Point Estimate: A single value used to estimate a parameter.

2.1.1. Estimator and Estimates

Any sample statistic used to estimate or measure a population parameter is called an estimator, that is, an estimator is a sample statistic used to estimate a population parameter. The sample mean $\bar{x}$ can be an estimator of the population mean μ , and the sample proportion can be used as an estimator of the population proportion P .

Statistical Estimation refers the *procedure* of using a sample statistic (measure) to estimate a population parameter.

2.1.2. Types of Statistical Estimates

We can make two types of estimates about a population: a point estimate and an interval estimate.

- A. **Point estimate:** It is a single number that is used to estimate an unknown population parameter. Point estimate is the values computed from sample distribution that is used to estimate the population parameter. The sample mean, $\bar{x}$ is a point estimator for the population mean, μ ; and sample proportion, $\bar{p}$ estimates parameter population proportion, P .
- B. **Interval estimate:** It is a range of values used to estimate the population parameter. It places the unknown population parameter between two limits. It has ranges to estimate the population. It also assumes or considers the errors associated with the sampling procedure. It indicates the errors in two ways: by the extent of its range and by the probability of the true population parameter lying within that range.

Example: The mean of the age of men attending a show is between 28 and 36 years or it can be written as $28 \leq \text{age} \leq 36$.

2.1.3. **Criteria of a Good Estimator**

There are three criteria developed to compare statistical estimators in terms of their worth as an estimator:

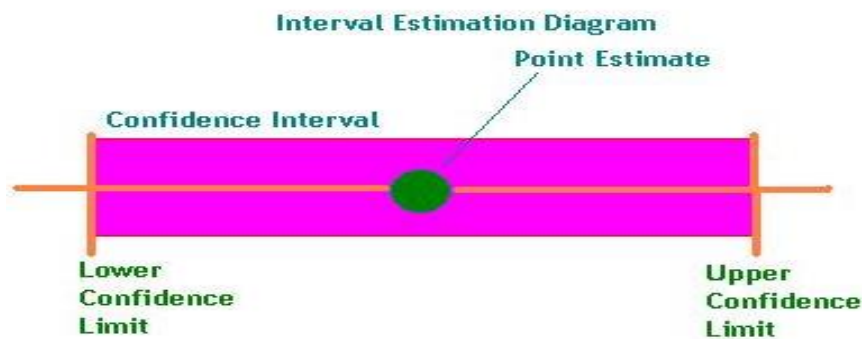
- 1. **Unbiasedness:** An unbiased estimator is a statistic that has an expected value equal to the population parameter being estimated.
Examples: The sample mean, $\bar{x}$ is an unbiased estimator of the population mean, μ .
- 2. **Efficiency:** Efficiency refers to the size of the standard error or standard deviation of the statistic. An efficient estimator considers the reliability of the estimator in terms of its tendency to have a smaller standard error for the same sample size when compared each other.
- 3. **Consistency:** A statistic is a consistent estimator of a population parameter if as the sample size increases; it becomes almost certain that the value of the statistic comes very close to the value of the population parameter. If an estimator is consistent, it becomes more reliable with large samples. The standard error of a consistent estimator becomes smaller as the sample size gets larger.

The sample mean and sample proportions are consistent estimators, since from their formulas as n get big, the standard error becomes small, that is, $\sigma \bar{x} = \frac{\sigma}{\sqrt{n}}$ and $\sigma \bar{p} = \sqrt{\frac{pq}{n}}$.

2.1.Interval Estimates and Confidence Intervals

The probability that we associate with an interval estimate is called the confidence level. This probability indicates how confident we are that the interval estimate will include the population parameter. A higher probability means more confident.

Interval Estimation establishes an interval consisting of a lower limit and an upper limit in which the true value of the population parameter is expected to fall. This interval is called "*Confidence Interval*" in the parlance of inferential statistics.



2.2.1 Interval Estimation of a Population Mean

A. Large sample Case ($n \geq 30$) and Standard Deviation is Known

The z-score for the normal variable statistics is used to help the determination of the interval endpoints that correspond to the probability of degree of certainty one which to use for the interval estimator.

An interval estimator for the mean is given by the following formula:

$$\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \text{ Or } \mu : \bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

For reasonably large samples, the results of the central limit theorem state the following:

1. 90% of the sample means selected from population will be within 1.64 standard deviations of the population mean μ .
2. 95% of the sample means selected from population will be within 1.96 standard deviations of the population mean μ .
3. 99% of the sample means will lie within 2.58 standard deviations of the population mean.

Example: Find an interval estimator of the sample mean of a random variable of sample size 49 if the population standard deviation is 5 and the sample mean is 15. Assume a 95% confidence interval for the population true mean.

Solution: $\sigma_{\bar{X}} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}} = \frac{5}{\sqrt{49}} = 0.7143$, and the given $\bar{x} = 15$.

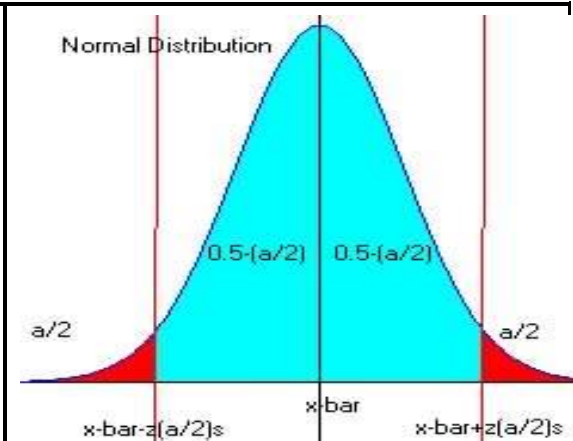
Since the level of significance, alpha is 5% (100 - 95) or 0.05, so $\alpha/2 = 0.025$. From the standard normal distribution reference table, $Z_{\alpha/2}$ appears at the endpoints of the normal distribution. At the probability of 0.025; thus, $Z_{0.025} = 1.96$.

So, the **95% confidence interval** for the example is $\mu : 15 \pm 1.96 \cdot 0.7143 = 15 \pm 1.4$

So Confidence Interval = $13.6 < \mu < 16.4$

The confidence interval $(1-\alpha)$ for the population mean, μ when the sample size is large ($n \geq 30$) is:

$\bar{x} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$ or the endpoints
 $\mu : \bar{x} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$



The **maximum error of the estimate**, E , with level of confidence $1-\alpha$, is the error associated with the estimate of the population mean from the sample mean and is given by the formula below:

$E = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$ or The Confidence Interval, when both the mean and the standard deviations are estimated from the sample mean and variance if $\bar{x} \pm E$ or as above

$\mu : \bar{x} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

Example: Find the 99% confidence interval estimate of the true population mean income if a sample of 100 families gives a sample mean of \$28,500. From previous experience we know that the population standard deviation is \$5,000.

Solution: Using $\alpha = 1 - 0.99 = 0.01$, the probabilities are $\alpha/2 = 0.01/2 = 0.005$; thus, $Z_{0.005} = 2.57$

So our CI estimate is: $\mu : \bar{x} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = \$28,500 \pm 2.57 \cdot \frac{\$5,000}{\sqrt{100}} = \$28,500 \pm \$1,285$

$\$27,215 \leq \mu \leq \$29,785$

B. Small Sample Case ($n < 30$) with Unknown Standard Deviation

In the case of small sample size ($n < 30$) and with unknown standard deviation, t distribution is applied.

$\bar{X} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$, where; $1-\alpha$ is the confidence coefficient.

- $t_{\alpha/2}$ is the t value providing an area of $\alpha/2$ in the upper tail of a t distribution with $n-1$ degree of freedom.
- S is the sample standard deviation.
- The population is assumed to have normal probability distribution.

I. Using the t Distribution to find Interval estimations

The t distribution is commonly called student's t distribution, or simply student's distribution. Use of the t distribution for estimating is required whenever the sample size is less than 30 and the population standard deviation not known. Furthermore, in using the t distribution, we assume that the population is normal or approximately normal.

II. Characteristics of the t Distribution

The t distribution shares some characteristics of the normal distribution and differs from it in others. The t distribution is similar to the standard normal distribution in these ways:

1. It is bell-shaped.
2. It is symmetric about the mean.
3. The mean, median, and mode are equal to 0 and are located at the center of the distribution.
4. The curve never touches the x axis.

The t distribution differs from the standard normal distribution in the following ways:

1. The variance is greater than 1.
2. The t distribution is actually a family of curves based on the concept of *degrees of freedom*, which is related to sample size.
3. As the sample size increases, the t distribution approaches the standard normal distribution.

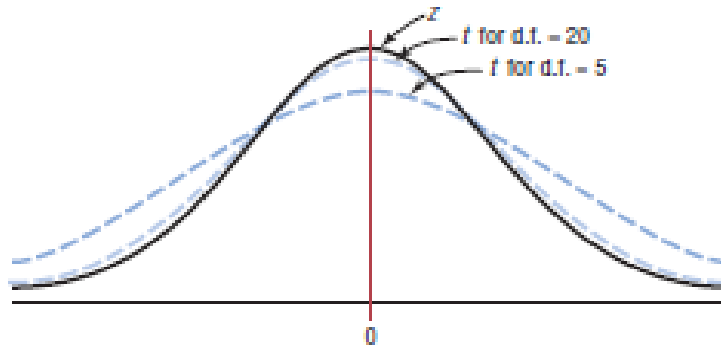


Figure. Normal distribution, t distribution for sample size $n=21$, and t distribution for sample size $n=6$.

III. Degrees of Freedom

Many statistical distributions use the concept of degrees of freedom, and the formulas for finding the degrees of freedom vary for different statistical tests. The **degrees of freedom** are the number of values that are free to vary after a sample statistic has been computed, and they tell the researcher which specific curve to use when a distribution consists of a family of curves.

For example, if the mean of 5 values is 10, then 4 of the 5 values are free to vary. But once 4 values are selected, the fifth value must be a specific number to get a sum of 50, since $50/5 = 10$. Hence, the degrees of freedom are $5 - 1 = 4$, and this value tells the researcher which t curve to use.

Degree of freedom = $n-1$

Example 1: An agricultural chemical retail firm wants to estimate the average number of gallons of weed killer sold per day for the purpose of accurately forecasting and controlling inventory. Twelve business days were monitored, and average daily sales of 10 gallons were recorded. The sample yielded standard deviation of 2 gallons. Calculate the confidence limits at the 95% level?

Solution:

$n = 12$	$\bar{X} = 10$	$s = 2 \quad CL = 0.95$
$\delta = 1 - CL = 1 - 0.95 = 0.05$		$df = n - 1 = 12 - 1 = 11$
$\delta/2 = 0.05/2 = 0.025$		$t_{\delta/2} = t_{0.025} \text{ at } df (11) = 2.201$

Confidence interval:

$$\begin{aligned} \bar{X} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} &= 10 \pm 2.201(2/\sqrt{12}) = 10 \pm 2.201(0.57735) = 10 \pm 1.27 \\ &= (8.73, 11.27) \end{aligned}$$

Example 2: A drug company is testing a new drug which is supposed to reduce blood pressure. From the six people who are used as subjects, it is found that the average drop in blood pressure is 2.28 points, with a standard deviation of .95 points. What is the 95% confidence interval for the mean change in pressure?

Solution:

$$\begin{aligned}\bar{X} &= 2.28, \quad S = 0.95, \quad 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05, \quad \alpha/2 = 0.025 \\ &\Rightarrow t_{\alpha/2} = 2.571 \text{ with } df = 5 \text{ from table.} \\ \Rightarrow \text{The required interval will be } \bar{X} \pm t_{\alpha/2} S / \sqrt{n} \\ &= 2.28 \pm 2.571 * 0.95 / \sqrt{6} \\ &= 2.28 \pm 1.008 \\ &= \underline{\underline{(1.28, 3.28)}}\end{aligned}$$

4.2.2. Interval Estimation for a Population Proportion

Samples are often used to estimate a proportion of occurrences in a population. For example, the government estimates by a sampling procedure the unemployment rate, or the proportion of unemployed people, in the country workforce.

Let's express the proportion of successes in a sample by $\bar{p}$.

Mean of the sampling distribution of the proportion is: $\mu_{\bar{p}} = p$

For population $P \pm Z \sigma_p$

For sample $\bar{p} \pm Z \sigma_{\bar{p}}$

Standard error of the proportion is: $\sigma_{\bar{p}} = \sqrt{\frac{pq}{n}}$.

Confidence limits are: $\bar{p} \pm Z \sqrt{\frac{pq}{n}}$

Example: When a sample of 70 retail executives was surveyed regarding the poor performance of the retail industry, 66% believed that decreased sales were due to unseasonably warm temperatures, resulting in consumers' delaying purchase of cold-weather items.

- Estimate the standard error of the proportion of retail executives who blame warm weather for low sales.
- Find the upper and lower confidence limits for this proportion, given a 95% confidence level.

Solution: $n = 70$, and $\bar{p} = 0.66$

$$a) \sigma \bar{p} = \sqrt{\frac{pq}{n}} = \sqrt{\frac{0.66(0.34)}{70}} = 0.0566$$

$$b) \bar{p} \pm Z \sqrt{\frac{pq}{n}} = 0.66 \pm 1.96(0.0566) = 0.66 \pm 0.111 = [0.316, 0.77].$$

Exercises

1. The operation manager of a certain Tele-center is in the process of developing an operation plan. For that purpose, he takes a random sample of 60 calls from the company records and finds that the mean sample length for a call is 4.26 minutes. Past history for these types of calls has shown that the population standard deviation for call length is about 1.1 minutes. Assuming that the population is normally distributed and he wants to have a 95% confidence, help him in estimating the population mean.
2. A survey conducted by a CSA found that the sample mean age of men was 44 years and the sample mean age of women was 47 years. Altogether, 454 people from Oromia were included in the reader poll -340 women and 114 men. Assume that the population standard deviation of age for both men and women is 8 years. Develop a 99% confidence interval estimate for the mean age of the population men.
3. Suppose that a survey is being conducted in a company that has 800 workers. A random sample of 50 of these workers reveals that the average sample age is 34.3 years, and the sample standard deviation is 8 years. Assuming normality, construct a 98% confidence interval to estimate the average age of all workers in this company.
4. A recent study showed that the modern working person experiences an average of 2.1 hours per day of distractions (phone calls, e-mails, impromptu visits, etc.). A random sample of 50 workers for a large corporation found that these workers were distracted an average of 1.8 hours per day and the population standard deviation was 20 minutes. Estimate the true mean population distraction time with 90% confidence, and compare your answer to the results of the study.
5. A survey of 30 emergency room patients found that the average waiting time for treatment was 174.3 minutes. Assuming that the population standard deviation is 46.5 minutes, find the best point estimate of the population mean and the 99% confidence of the population mean.
6. Ten randomly selected people were asked how long they slept at night. The mean time was 7.1 hours, and the standard deviation was 0.78 hour. Find the

95% confidence interval of the mean time. Assume the variable is normally distributed.

7. If a random sample of 27 items produces a mean of 128.4 and standard deviation of 20.6 what is the 98% confidence interval for μ ? Assume that x is normally distributed for the population. What is the point estimate?
8. A meteorologist who sampled 13 thunderstorms found that the average speed at which they traveled across a certain state was 15 miles per hour. The standard deviation of the sample was 1.7 miles per hour. Find the 99% confidence interval of the mean. If a meteorologist wanted to use the highest speed to predict the times it would take storms to travel across the state in order to issue warnings, what figure would she likely use? 13.6 _ m _ 16.4; 16.4 miles per hour
9. A recent study of 28 employees of XYZ Company showed that the mean of the distance they traveled to work was 14.3 miles. The standard deviation of the sample mean was 2 miles. Find the 95% confidence interval of the true mean. If a manager wanted to be sure that most of his employees would not be late, how much time would he suggest they allow for the commute if the average speed were 30 miles per hour?
10. For a group of 22 college football players, the mean heart rate after a morning workout session was 86 beats per minute, and the standard deviation was 5. Find the 90% confidence interval of the true mean for all college football players after a workout session.
11. The national average for the number of students per teacher for all U.S. public schools is 15.9. A random sample of 12 school districts from a moderately populated area showed that the mean number of students per teacher was 19.2 with a variance of 4.41. Estimate the true mean number of students per teacher with 95% confidence. How does your estimate compare with the national average?
12. A gasoline service station shows a standard deviation of Birr 6.25 for the changes made by the credit card customers. Assume that the station's management would like to estimate the population mean gasoline bill for its credit card customers to be within an error of Birr 1.00. For a 95% confidence level, how large a sample would be necessary?
13. A random sample of shoppers at a convenience store is selected to see how much they spent on that visit. The standard deviation of the population is \$6.43. How large a sample must be selected if the researcher wants to be 99% confident of finding whether the true mean differs from the sample mean by \$1.50?

14. A random sample of 205 college students were asked if they believed that places could be haunted, and 65 responded yes. Estimate the true proportion of college students who believe in the possibility of haunted places with 99% confidence. According to *Time* magazine, 37% of Americans believe that places can be haunted.
15. A survey conducted by Sallie Mae and Gallup of 1404 respondents found that 323 students paid for their education by student loans. Find the 90% confidence of the true proportion of students who paid for their education by student loans.
16. The national average for the percentage of high school graduates taking the SAT is 49%, but the state averages vary from a low of 4% to a high of 92%. A random sample of 300 graduating high school seniors was polled across a particular state area, and it was found that 195 had taken the SAT. Estimate the true proportion of high school graduates in this region who take the SAT with 95% confidence.
17. A researcher wishes to estimate, with 95% confidence, the proportion of people who own a home computer. A previous study shows that 40% of those interviewed had a computer at home. The researcher wishes to be accurate within 2% of the true proportion. Find the minimum sample size necessary.
18. It is believed that 25% of U.S. homes have a direct satellite television receiver. How large a sample is necessary to estimate the true population of homes which do with 95% confidence and within 3 percentage points? How large a sample is necessary if nothing is known about the proportion?
19. America's young people are heavy Internet users; 87% of Americans ages 12 to 17 are Internet users (*The Cincinnati Enquirer*, February 7, 2006). Myspace was voted the most popular website by 9% in a sample survey of Internet users in this age group. Suppose 1400 youths participated in the survey. What is the margin of error, and what is the interval estimate of the population proportion for which Myspace is the most popular website? Use a 95% confidence level.